

Two factor ANOVA model (under balanced design with replication)

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LSM

Response Y

Categorical predictors — factor A with a levels (categories)

factor B with b levels

$a, b \geq 2$

Under randomized experiment / observational study

either complete randomized design (CRD)
a block randomized design where the blocking
is wrt the observational factor(s)

In the drug development study, drug type is an experimental factor, and gender is an observational factor.

Model:

$$\gamma_{ijk} = \mu_{ij} + \varepsilon_{ijk}$$

$$i = 1, \dots, a$$

$$j = 1, \dots, b$$

$$k = 1, \dots, m$$

balance)

$$= \underbrace{\mu_{..} + \alpha_i + \beta_j + \gamma_{ij}}_{\text{systematic}} + \epsilon_{ijk}$$

under identifiability constraints

$$\star \left\{ \begin{array}{l} \sum_{i=1}^a \alpha_i = 0, \sum_{j=1}^b \beta_j = 0, \sum_{j=1}^b v_{ij} = 0 \quad \forall i \\ \sum_{i=1}^a v_{ij} = 0 \quad \forall j \end{array} \right.$$

$$(E_{ijk} \stackrel{iid}{\sim} N(0, \sigma^2))$$

Estimation of the parameter

Find BLUE of $\{\mu_{..}, \{\alpha_i\}, \{\beta_j\}, \{\nu_{ij}\}\}$

Express all these parameters as linear function of $\{\mu_{ij}\}$

$$\mu_{..} = \frac{1}{ab} \sum_{i=1}^a \sum_{j=1}^b \mu_{ij} \quad \alpha_i = \frac{1}{b} \sum_{j=1}^b \mu_{ij} - \mu_{..}$$

$$\beta_j = \frac{1}{a} \sum_{i=1}^a \mu_{ij} - \mu_{..} \quad \nu_{ij} = \mu_{ij} - \alpha_i - \beta_j - \mu_{..}$$

BLUE for this parameter will be obtained by replacing μ_{ij} by $\hat{\mu}_{ij} = \bar{Y}_{ij}$.

LSE for $\{\mu, \{\alpha_i\}, \{\beta_j\}, \{\nu_{ij}\}\}$ ($\mu = \mu_{..}$)

$$Q(\mu, \{\alpha_i\}, \{\beta_j\}, \{\nu_{ij}\}) = \sum_i \sum_j \sum_k (Y_{ijk} - \mu - \alpha_i - \beta_j - \nu_{ij})^2$$

↪ Minimize under constraints (*)

BLUE $\stackrel{?}{=} \text{LSE}$

Let $\hat{\mu}, \{\hat{\alpha}_i\}, \{\hat{\beta}_j\}, \{\hat{\nu}_{ij}\}$ be the BLUE for $\{\mu, \{\alpha_i\}, \{\beta_j\}, \{\nu_{ij}\}\}$. Then notice that these estimates satisfy the constraints.

$$\hat{\mu} = \bar{Y}_{...}, \hat{\alpha}_i = \bar{Y}_{i..} - \bar{Y}_{...}, \hat{\beta}_j = \bar{Y}_{.j.} - \bar{Y}_{...}$$

$$\hat{\gamma}_{ij} = \bar{Y}_{ij.} - \bar{Y}_{i..} - \bar{Y}_{.j.} + \bar{Y}_{...}$$

$$\hat{\mu} + \hat{\alpha}_i + \hat{\beta}_j + \hat{\gamma}_{ij}$$

$$G(\mu, \{\alpha_i\}, \{\beta_j\}, \{\gamma_{ij}\})$$

$$= \bar{Y}_{ij.} = \hat{\mu}_{ij}$$

$$= \sum_i \sum_j \sum_k \left[(Y_{ijk} - \bar{Y}_{ij.}) + (\hat{\mu} - \mu) + (\hat{\alpha}_i - \alpha_i) + (\hat{\beta}_j - \beta_j) + (\hat{\gamma}_{ij} - \gamma_{ij}) \right]^2$$

$$= \sum_i \sum_j \sum_k \left\{ (Y_{ijk} - \bar{Y}_{ij.})^2 + (\hat{\mu} - \mu)^2 + (\hat{\alpha}_i - \alpha_i)^2 + (\hat{\beta}_j - \beta_j)^2 + (\hat{\gamma}_{ij} - \gamma_{ij})^2 \right\}$$

$$= \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{ij.})^2 + mab(\hat{\mu} - \mu)^2 + mb \sum_i (\hat{\alpha}_i - \alpha_i)^2$$

$$+ ma \sum_j (\hat{\beta}_j - \beta_j)^2 + m \sum_i \sum_j (\hat{\gamma}_{ij} - \gamma_{ij})^2$$

$$= \tilde{Q}(\mu, \{\alpha_i\}, \{\beta_j\}, \{\gamma_{ij}\})$$

↑ under $\star$ (unconstrained minimizer of $\tilde{G}$ satisfies $\star$ and $G = \tilde{G}$ under the constraint) $\square$

Byproduct: ANOVA decomposition

$$SSTO = \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{ij.})^2$$

$$SSR = \sum_i \sum_j \sum_k (Y_{ijk} - \bar{Y}_{ij.})^2$$

Set $\mu = \bar{Y}_{...}$, $\alpha_i = 0$, $\beta_j = 0$, $\gamma_{ij} = 0$ satisfy (*)
on the LHS

$$\Rightarrow SSTO = SSE + \underbrace{mb \sum_i \hat{\alpha}_i^2}_{SSA} + \underbrace{ma \sum_j \hat{\beta}_j^2}_{SSB} + \underbrace{m \sum_i \sum_j \hat{\gamma}_{ij}^2}_{SSAB}$$

Interpretation of terms in the ANOVA decomposition

$$SSA = mb \sum_i \hat{\alpha}_i^2$$

$$SSB = ma \sum_j \hat{\beta}_j^2$$

$$SSAB = m \sum_i \sum_j \hat{\gamma}_{ij}^2$$

$$E(SSA) = mb \sum_i \alpha_i^2 + (a-1) \sigma^2$$

$$E(SSB) = ma \sum_j \beta_j^2 + (b-1) \sigma^2$$

$$\hat{\alpha}_i = \bar{Y}_{i..} - \bar{Y}_{...}$$

$$E \hat{\alpha}_i = \alpha_i = \frac{1}{b} \sum_j \mu_{ij} - \mu_{..}$$

$$\text{Var}(\hat{\alpha}_i) = \text{Var}\left(\bar{Y}_{i..} - \frac{1}{a} \sum_{i'=1}^a \bar{Y}_{i'..}\right)$$

$$= \left(1 - \frac{1}{a}\right)^2 \text{Var}(\bar{Y}_{i..}) + \frac{1}{a^2} \sum_{i' \neq i} \text{Var}(\bar{Y}_{i'..})$$

$$= \left(1 - \frac{1}{a}\right)^2 \frac{\sigma^2}{mb} + \frac{a-1}{a^2} \frac{\sigma^2}{mb}$$

$$= \frac{(a-1)^2 + (a-1)}{a^2} \frac{\sigma^2}{mb}$$

$$= \frac{a-1}{a} \frac{\sigma^2}{mb}$$

$$\frac{SSA}{a-1} = MSA \quad \begin{pmatrix} \hat{\alpha}_1 \\ \vdots \\ \hat{\alpha}_a \end{pmatrix}$$

df (a-1) ←

One linear constraint

$$MSE = \frac{SSE}{mab - ab}$$

$$= \frac{SSE}{(m-1)ab}$$